Meg Montañez Davenport

BLOCKCHAIN HEALTH SYSTEMS ENGINEER
WEB3 ARCHITECT · APPLIED RESEARCHER

CONTACT

- future.systems.lab@proton.me
- sovereigncoverage@futuresystemslab.io
- NC, USA — Remote
- linkedin.com/in/dr-meg-montanez-davenport
- github.com/Future-Systems-Lab/Future-Systems-Lab-profile
- americaoutloud.news/author/meg-montanez-davenport

BLOCKCHAIN & WEB3

Solidity · Hardhat · Remix IDE · ERC-20/721 · Brave Wallet · EIP-6963 · window.ethereum · DIDs / SSI · IPFS · Filecoin · Merkle DAGs · ZK Proofs · libp2p · Blockscout

ENGINEERING & DATA

Next.js · PostgreSQL 17 · Python · Flask · Streamlit · systemd · REST APIs · Vercel · SSH Ed25519 · Git CI/CD · Agile/Scrum · Tableau

HEALTHCARE & INFORMATICS

HL7 · FHIR · OMOP · HIPAA · EHR Mapping · Clinical Data Models · Orthomolecular

PRACTICE MANAGEMENT SYSTEMS

Open Dental · SoftDent · Dentrix · Denticon · Archy · Eaglesoft · MacPractice

ORGANIZATIONS

- INCOSE — Systems Engineering
- AMIA — Medical Informatics
- HIMSS — Health IT
- ISC² — Cybersecurity
- PDMA — Product Management
- ISOM — Orthomolecular Medicine
- NANP — Nutrition Professionals

CERTIFICATIONS

Certified Blockchain Healthcare Professional (CBHP) — Blockchain Council · Apr 2025

Certified Smart Contract Auditor Blockchain Council · Jan 2026

SUMMARY

I design and deploy compliance-aware decentralized systems — using regulated mental healthcare as a proving ground for architectures that scale across complex domains. As founder of Future Systems Lab, I move from concept to production: on-chain deployments, FHIR-aligned consent flows, and governance-aware infrastructure built to HIPAA standards from day one. My background spans naturopathic psychology and blockchain product architecture — I think in systems, across disciplines, at the intersection of governance and user experience. Eight Ethereum smart contracts deployed and verified on Sepolia — every FSL system is in production. Pursuing a Doctor of Engineering to formalize applied research methodology and advance systems leadership at the intersection of blockchain, AI, and decentralized governance.

ENGINEERING EXPERIENCE

Lead Engineer — SovereignLedger Sovereign Health Governance Infrastructure

2026 – Present

Sepolia: 0x4afA577fA914068451e0Aa97b61F23960f02aCc4

- Engineered a full PostgreSQL session governance schema spanning participants, sovereign guides, sessions, attestations, documents, audit logs, consent grants, and compliance log tables.
- Built Phase 3 SMS-based approval-gated automation with case-insensitive approve/deny/defer, 4-hour timeout, and max-defer logic — deployed to production VPS via systemd.
- Architected SovereignLedger as a standalone blockchain governance layer for session governance, attestation, and participant-controlled record verification — integrates with any payment pipeline (Crypto → XRP → Sovereign Navigation) and deployable independently of any single platform.

Production Architecture Lead — Riccobene Family Dentistry

2019 – 2025

70+ office network, North Carolina

- Designed and deployed two-tier production-share compensation model adopted company-wide across Riccobene's 70+ office network — 27% standard tier (network-wide rollout) and 33% architect tier (system designer compensation).
- Engineered restructured workflow processes during pandemic-driven hygienist shortage; documented hygienist production grew from $380,710 in 2021 to $624,000 annualized through October 2024 — 64% increase across four documented years of continuous year-over-year growth.
- Hygienist share of total office production grew from 30.4% to 38.1% — system redesign enhanced overall office economics rather than redistributing existing revenue.
- Recruited directly by CEO Michael Riccobene for C-suite engagement; maintained 6-year engagement while concurrently operating wellness businesses and founding Future Systems Lab.

Open-Source Case Study Lead — AlchemistForge / mental-health-on-chain

2026 – Present

- Scaffolded mental-health-on-chain as a five-entry public smart contract awareness series; deployed 24/7 campaign orchestration across Dev.to, Twitter, Discord, GitHub, and Telegram.
- Recorded on-chain case study as Participant #1 with Transmuted and Celebrated events confirmed at wallet 0xf22cbF25deEeA36FFF828BAf73CCb049345eF248.

Founder & Systems Architect — Future Systems Lab

2023 – Present

github.com/Future-Systems-Lab/Future-Systems-Lab-profile

- Deployed and verified eight Ethereum smart contracts on Sepolia testnet: HypnoNeuroToken, EHTv2, MindMasteryNFT, SovereignLedger, AlchemistForge, BenevolenceFund, SovereignAchievement, and HNT v2.

Health Informatics — Johns Hopkins May 2025

Web3 & Blockchain — INSEAD Apr 2025

CISA — Packt Aug 2025

SSI Master — DizmeID Mar 2025

Medical Neuroscience — Duke SOM Apr 2025

Health Informatics on FHIR — Georgia Tech Mar 2025

Business Analyst — IBM Jun 2025

Data Analytics · UX Design · PM — Google Jun 2025

IPFS · Filecoin · libp2p · Merkle DAGsIPFS / Filecoin Foundation / ProtoSchool · 2025

NC Accident & Health Insurance Producer Pre-licensing completed Apr 2026, exam scheduled

National Provider Identifier (NPI): 1497696264 Naturopath (175F00000X), NPPES, 2026

- Architected a 17-agent Master Command Center spanning Audit, Smart Contract, Backend Build, Frontend/UI, Compliance, Security, Testing/QA, and Research agents with full dropdown panel orchestration.
- Deployed production VPS automation stack (clawdbot) on OpenClaw VPS: Flask approval server, Telegram bot, Streamlit analytics dashboard, and two systemd services running continuously.
- Engineered EncryptAI — a 19-step canonical monorepo (frontend/, backend/, labs/, contracts/, game/) with Brave Wallet authentication via window.ethereum + EIP-191, PostgreSQL 17, and blockchain writes via viem — Step 2 live on Vercel.
- Established SSH-authenticated signed commits, GitHub org governance, and Rights Reserved licensing policy across all FSL repositories.

Healthcare Data Architect — HypnoNeuro Mental Wellness Platform

2023 — Present

HypnoNeuro · EncryptHealth · SovereignLedger · NeuroBalance Watch — modular systems engineered to operate independently or as an integrated ecosystem

- Shipped 27 browser-based therapeutic wellness games across three progressive treatment levels (L1 Hypnosis, L2 Orthomolecular, L3 Inner-Child Healing) — fully live in production with Brave Wallet gating, HNT token rewards, and NFT-based tier access.
- Architected dual-chain healthcare session anchoring — every session cryptographically committed to both Ethereum Sepolia and XRPL testnet, generating a tamper-evident audit trail for sovereign health records.
- Built ISOM-aligned 4-tier provider credentialing system (Platinum/Gold/Silver/Verified) with automated license verification via Verifiable.com and AI-assisted credential review using Claude Vision.
- Implemented Walsh Protocol AI recommendation engine — biotype classification across 5 neurotransmitter systems (Trp, Tyr, Cortisol, Dopamine, Epi/NorEpi) with plant-based nutrient protocol generation.
- Deployed avatar-based anonymous session rooms via Daily.co WebRTC — wallet-connected, provider-facilitated video sessions with sessionStorage-persisted character customization and SovereignLedger billing anchor.
- Designed white-label operator licensing model with three commercial tiers (Starter $497/mo, Growth $1,497/mo, Enterprise custom) — full registration flow, Telegram-notified applications pipeline, and operator config abstraction layer.
- Completed 35-step canonical build across 7 sprints — all five platforms live on Vercel and interconnected: HypnoNeuro, EncryptHealth, SovereignLedger, NeuroBalance, Provider Portal, and FSL Command Center.
- **Sprint 1–3:** Core platform buildout — 27 therapeutic games, Brave Wallet + XRPL dual-wallet auth, PostgreSQL schema, FHIR R4 export, XRPL session anchoring, NFT gating, HNT token rewards, provider portal backend.
- **Sprint 4–5:** Avatar Session Rooms (Daily.co WebRTC), AlchemistForge HNT bridge, endocannabinoid baseline ML scorer, COA Lookup, Support Circles, Walsh Protocol recommendation engine, wellness scoring, XRPL wallet connect.
- **Sprint 6:** XRPL wallet polish (GemWallet/Crossmark/Xaman/manual), demo mode with watermark and seeded leaderboard, biosensor infrastructure spec (5 target hardware partners), FSL Command Center on Vercel, Telegram operator notification pipeline.
- **Sprint 7 (current):** Intake form Walsh Protocol biotype pre-screening, win screen confetti + NFT progress tracking + share flow, support circle categories and HNT bonus, ecosystem health monitoring (cron + Telegram alerts), OSS portfolio formalization.
- Planned hardware integration: NeuroBalance biosensor layer targeting Nutromics (sweat biomarkers), Polar (HRV), OpenBCI (EEG), Emotiv, and Dexcom (CGM) — full API spec and device schema documented.

Founder & Product Developer — SunCodes Holistic Health

2018 — 2024

- Developed a seed-to-shelf CBD product line in an unregulated market — managed full product lifecycle including formulation, third-party COA testing, GMP documentation, labeling, barcode integration, and retail packaging compliance.
- Led full renovation and interior redesign of a historic downtown smoke shop, converting it into a branded therapeutic wellness center — managed contractor relationships, spatial design, and retail buildout from concept to opening.

- Founded and operated a clinical wellness platform applying naturopathic psychology and orthomolecular frameworks to mental health and trauma recovery, with digital compliance tracking and personalized client protocol systems.

OPEN SOURCE CONTRIBUTIONS

Open Source Contributor — Healthcare & Blockchain

Jul 2025 — Present

Self-Directed / Independent

- **Ethereum / EIPs #11409** — EIP document structure formatting fix · Open · Connects to FSLToken/HypnoNeuroNFT EIP-20/721 standard compliance
- **OpenZeppelin Contracts #6414** — ReentrancyGuard documentation clarification · Open · FSL contracts inherit from OpenZeppelin audited base contracts
- **Uniswap v3-core #1077** — Invariant test note for pool math · Open · Reference for future HNT/ETH liquidity pool design
- **Hyperledger Fabric #5419** — Build and runtime environment notes · Open · Permissioned ledger patterns for FSL provider credentialing
- **OpenMRS / FHIR2 #594** — Quickstart pointer for healthcare interoperability · Closed · Same FHIR2 module stack as EncryptHealth FHIR R4 export
- **BigchainDB #2763** — Python quickstart documentation · Open · Immutable data patterns informing NeuroBalance biosensor anchoring
- **Balancer v2 #2648** — Pool math reference documentation · Open · HNT liquidity pool and AlchemistForge bridge architecture reference
- **DIF / Presentation-Exchange #486** — OIDC4VP consent example · Open · SovereignLedger decentralized identity and provider credentialing flows

Applied projects:

- **Secure Health Login** — practitioner-facing smart wallet authentication app implementing EIP-712, DID/OIDC, and HIPAA-aligned access control — live on Vercel

PUBLICATIONS

> **"Why Your Medical Record Belongs in Your Digital Wallet"**
> America Out Loud · April 2026 · americaoutloud.news
> Argued that sovereign wallet-based health data ownership eliminates institutional gatekeeping — patients should control their medical records the same way they control their financial assets.

> **"Reprogram Your Limiting Beliefs on the Blockchain While Learning Web3"**
> America Out Loud · April 2026 · americaoutloud.news
> Introduced AlchemistForge as Case Study #1 of the mental-health-on-chain series — demonstrating how Jungian shadow integration can be permanently recorded on the Ethereum blockchain.

> **"Ending the Insurance Verification Nightmare"**
> America Out Loud · March 1, 2026 · americaoutloud.news
> Argued that blockchain governance resolves systemic insurance fraud — duplicate claims, identity misuse, phantom billing — through decentralized eligibility verification and immutable audit trails.

> **"Blockchain Consent Transforms Patient Data Control"**
> America Out Loud · March 2026 · americaoutloud.news
> Examined how blockchain-based consent architecture returns data sovereignty to patients — arguing that smart contract-governed consent flows resolve the systemic failures of centralized health data management and establish a verifiable, patient-controlled alternative.

EDUCATION

BS, Management Information Systems — In Progress
CSU Global — Regionally Accredited (HLC)

Doctor of Naturopathic Psychology — Conferred
Kingdom College of Natural Health

ENTITY & INTELLECTUAL PROPERTY

Future Systems Lab LLC (Wyoming) — EIN 42-2050630

USPTO Trademark "FUTURE SYSTEMS LAB" — Serial No. 99533250, Class 42 (filed Dec 2025; Office Action response filed Apr 2026, examiner review pending)

National Provider Identifier (NPI): 1497696264 — Naturopath (175F00000X), NPPES, 2026

PROPOSED APPLIED RESEARCH

Decentralized Sovereign Health Governance: Engineering a privacy-first, blockchain-secured session governance system for sovereign wellness — advancing SovereignLedger from testnet deployment into a rigorous applied research framework examining smart contract-governed consent architecture, HIPAA-adjacent compliance engineering at the contract layer, and interoperability with HL7/FHIR standards at scale. | SovereignLedger on Blockscout